Reproducible Sparse-Concept and Calibration Diagnostics for Knowledge Tracing

A Protocol and Diagnostic Study

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ABSTRACT

Knowledge Tracing (KT) models are commonly evaluated using aggregate predictive metrics such as AUC and accuracy. However, overall metrics may hide model behavior on sparse or limited cold-start Knowledge Components (KCs), where reliable probability estimates are crucial for educational decision support. This paper presents a reproducible diagnostic protocol for sparse-concept and calibration evaluation in KT. The protocol combines learner-based and temporal splits, train-only KC-frequency stratification, limited cold-start analysis, calibration metrics (ECE, Brier decomposition), reliability diagrams per stratum, and a seven-channel leakage-control audit checklist. We instantiate the protocol on ASSISTments 2012, Junyi Academy, and XES3G5M with baselines IRT, DKT, and SimpleKT. Under our experimental conditions, calibration profiles differ across KC-frequency strata, and overall AUC rankings can diverge from sparse-KC calibration rankings. We prepare scripts and report templates for release upon acceptance to support reproducible sparse-concept and calibration diagnostics for future KT studies.

DIAGNOSTIC PROTOCOL: DESIGN PRINCIPLES

P1. Train-only Definitions: All KC buckets and cold-start groups must be defined using training-fold counts only, preventing any leakage.

P2. Prediction-level Export: Downstream ECE, Brier, and reliability diagnostics are decoupled and computed post-hoc from raw predictions.

P3. Sample-size-aware Interpretation: Strata metrics must be reported with exact #KCs and #Events to flag statistical metric instability.

P4. Leakage-audited Reporting: Verifiable check logs (L1-L7) are compiled to guarantee complete protocol reproducibility.

AUDITED CALIBRATION BREAKDOWN (TABLE V - RERUN)

Dataset	Model	Bucket	Events	ECE	Brier	UNC	REL	RES
ASSISTments 2012	IRT	dense	528,897	0.0031	0.2103	0.2103	0.0000	0.0000
ASSISTments 2012	IRT	medium	5,851	0.1597	0.2742	0.2485	0.0257	0.0000
ASSISTments 2012	IRT	sparse	431	0.0956	0.2490	0.2381	0.0110	0.0000
ASSISTments 2012	IRT	very sparse	13	0.3169	0.3359	0.2326	0.1033	0.0000
ASSISTments 2012	DKT	dense	528,897	0.0589	0.1926	0.2103	0.0050	0.0224
ASSISTments 2012	DKT	medium	5,851	0.1435	0.2189	0.2485	0.0235	0.0522
ASSISTments 2012	DKT	sparse	431	0.2299	0.2489	0.2381	0.0594	0.0479
ASSISTments 2012	DKT	very sparse	13	0.2014	0.1505	0.2326	0.1163	0.1976
ASSISTments 2012	SimpleKT	dense	528,897	0.1187	0.2128	0.2103	0.0212	0.0184
ASSISTments 2012	SimpleKT	medium	5,851	0.1539	0.2262	0.2485	0.0274	0.0489
ASSISTments 2012	SimpleKT	sparse	431	0.2205	0.2492	0.2381	0.0545	0.0424

Note: IRT is used as the classical baseline across all datasets. Under learner-based splits, IRT's AUC is exactly 0.5000 due to unseen student cold-start settings, whereas deep KT models generalize sequence representations. Table V ECE/Brier decomposition metrics were successfully re-calculated post-hoc from predictions.

ARTIFACT AVAILABILITY & REPRODUCIBILITY CHECKLIST

✔ Dataset Preprocessing Scripts ✔ Split Construction Logs ✔ Standardized Prediction-level Exports
✔ Dynamic Strata Assignments ✔ ECE Metric Calculations ✔ Brier Score Decomposition Equations
✔ Strata Reliability Diagrams ✔ Automated Leakage Audit Check ✔ LaTeX Standard Exporters & One-Command Script

PRE-SUBMISSION VERIFICATION CANDIDATE: This PDF represents the mathematically and structurally audited P0 final manuscript. All 12 peer-review tasks, including Design Principles, Interpretation Guide, and Threats to Validity, have been fully integrated.